ELEC3705 Fundamentals of Quantum Engineering

Laboratory 4 : Dynamical control of a qubit

Abstract

This lab explores how time-dependent Hamiltonians affect quantum state evolution, and the use of a rotating frame of reference to simplify our calculations using time-independent techniques.

Introduction

Time-Dependent Systems

We will now consider a Hamiltonian that is no longer static in time. Consider a single electron in a magnetic environment portrayed in Figure 1. The total magnetic field, $\mathbf{B_T}$ is given by

$$\mathbf{B_T} = \mathbf{B_x} + \mathbf{B_y} + \mathbf{B_z} \tag{1}$$

where

$$\mathbf{B_z} = B_z \vec{u}_z \tag{2}$$

$$\mathbf{B_x} = B_x \cos(\omega t) \vec{u}_x \tag{3}$$

$$\mathbf{B_y} = B_y \sin(\omega t) \vec{u}_y \tag{4}$$

and $\vec{u}$ is the unit vector in the specified direction and $B_x = B_y = B_\perp$ (B_z will be referred to as $B_\parallel$).

Note that the environment itself is time-dependent now which means the system Hamiltonian must also be time dependent. As in the previous lab, we can find our electron spin Hamiltonian to be

$$\begin{aligned}
\mathcal{H}(\omega) &= \frac{2\pi\,\gamma_e}{\hbar} \mathbf{B} \cdot \mathbf{S}\,[\mathrm{rad\,s^{-1}}] \\
&= \pi\,\gamma_e \begin{pmatrix} B_z & B_x\cos(\omega t) - B_y i \sin(\omega t) \\ B_x\cos(\omega t) + B_y i \sin(\omega t) & -B_z \end{pmatrix} \\
&= \pi\,\gamma_e \begin{pmatrix} B_\parallel & B_\perp e^{-i\omega t} \\ B_\perp e^{i\omega t} & -B_\parallel \end{pmatrix}
\end{aligned} \tag{5}$$

Question 1

What are the eigenvalues of the Hamiltonian described by equation (5)? Can you find the eigenvectors in terms of our standard computational basis (the eigenvectors of σ_z)?

The Rotating Reference Frame

As this Hamiltonian is periodic, we can imagine a frame of reference that rotates at the same rate as the Hamiltonian changes. This is called a rotating frame of reference, or just a rotating frame. To achieve this, we must redefine our basis state vectors to change as a function of time, instead of the Hamiltonian. We can transform our basis vectors from being time-independent to time-dependent as in equation (6). ω here is the same frequency as that of the perpendicular rotating magnetic field.

$$|\widetilde{\varphi_n}\rangle = \begin{pmatrix} e^{\frac{-i\omega t}{2}} & 0 \\ 0 & e^{\frac{i\omega t}{2}} \end{pmatrix} |\varphi_n\rangle \tag{6}$$

Which gives a new, time-independent system Hamiltonian [1]

$$\widetilde{\mathcal{H}} = \frac{\hbar}{2} \begin{pmatrix} -\Delta\omega & \omega_1 \\ \omega_1 & \Delta\omega \end{pmatrix} \tag{7}$$

where $\Delta\omega = \omega - \omega_0$ and $\omega_0 = \gamma_e B_\parallel$, $\omega_1 = \gamma_e B_\perp$.

Question 2

What happens to the system Hamiltonian when $\omega = \omega_0 = \gamma_e B_\parallel$? Qualitatively, how will the electron spin state evolve in time under this condition? Give your answer in both the rotating frame and the lab frame.

Now our system Hamiltonian, $\widetilde{\mathcal{H}}$, is time-independent, which means we have simplified our problem and should now examine how time evolution under a general time-independent Hamiltonian behaves.

[1] To find the new system Hamiltonian, we have to return to the Schrödinger equation with the new basis states. In summary, a change of basis yields a new Hamiltonian $\tilde{\mathcal{H}} = A\mathcal{H}A^\dagger + i\hbar\frac{\partial A}{\partial t}A^\dagger$, where A is the change of basis matrix given in equation (6). The second term of this formula is needed because we are changing to a non-inertial reference frame, which creates a 'fictitious energy'.

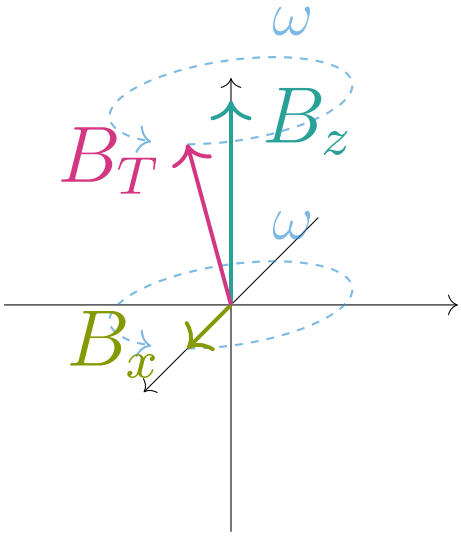


Figure 1: The time-dependent magnetic environment of our spin system.

Rabi Oscillations in a Time-Independent Hamiltonian

So far we have mainly considered diagonal matrices, but we can further describe what happens when we have off-diagonal elements in our Hamiltonian. Consider the Hamiltonian in equation (8), where ϵ is our detuning, and Δ is our coupling term.

$$\mathcal{H} = \begin{pmatrix} \epsilon & \Delta \\ \Delta & -\epsilon \end{pmatrix} \tag{8}$$

If we prepare $|\psi(t=0)\rangle = |\uparrow\rangle$, the state will evolve over time governed by equation (9).

$$P_{\downarrow}(t) = \Big|\, \langle\downarrow|\psi(t)\rangle \,\Big|^2 = \frac{\Delta^2}{\Delta^2+\epsilon^2} \sin^2\left(\frac{\sqrt{\Delta^2+\epsilon^2}}{\hbar} t\right) \tag{9}$$

We define the Rabi frequency, $\Omega_R = \dfrac{\sqrt{\Delta^2+\epsilon^2}}{\hbar}$.

Question 3

Derive Rabi's Formula given the Hamiltonian of equation (8), assuming $|\psi(t=0)\rangle = |\uparrow\rangle$. Consider both $\langle\downarrow|$ and $\langle\uparrow|$ for comparison with $|\psi(t)\rangle$.

Task

Consider an electron in a magnetic environment defined by

$$\mathbf{B_T} = \mathbf{B_x} + \mathbf{B_y} + \mathbf{B_z} \tag{10}$$

where

$$\mathbf{B_z} = 1\,\mathrm{T} \cdot \vec{u}_z \tag{11}$$

$$\mathbf{B_x} = 20\,\mathrm{mT} \cdot \cos((\omega + \Delta\omega)t)\vec{u}_x \tag{12}$$

$$\mathbf{B_y} = 20\,\mathrm{mT} \cdot \sin((\omega + \Delta\omega)t)\vec{u}_y \tag{13}$$

$$\omega = \omega_0 = 2\pi\gamma_e \cdot 1\,\mathrm{T} \approx 2\pi 28\,\mathrm{GHz} \tag{14}$$

Note the Larmor precession frequency due to B_z, ω_0. Now we are choosing the reference frame based on the Larmor precession, and so the frequency of the perpendicular rotating field is referred back to this with associated detuning $\Delta\omega$, which we can choose. The Hamiltonian we derived earlier in equation (7) still holds for this modified treatment. In reality, this would correspond to applying a known magnetic field (1 T) and applying a rotating magnetic field where you can control the frequency of rotation.

Question 4

What do the time evolution operators look like in the lab frame and the rotating frame?

Perform the time evolution in both the lab and rotating reference frames for the following given conditions. Also, for the evolution in the rotating frame, transform your eigenbasis back to the lab frame using equation (6) to verify that the lab frame and rotating frame give the same results. Calculate the Rabi frequency, Ω_R for each case.

a) $\Delta\omega = 2\pi \cdot 0\,\mathrm{Hz}$

b) $\Delta\omega = 2\pi \cdot 10\,\mathrm{MHz}$

c) $\Delta\omega = 2\pi \cdot 200\,\mathrm{MHz}$

Also plot the expectation values over time for each of S_x, S_y, S_z and therefore find the maximum probability of arriving in the $|\uparrow\rangle$ state, given an initial state of $|\downarrow\rangle$.

Extended Task

Consider now if $\mathbf{B_x} = 20\,\mathrm{mT} \cdot \cos((\omega + \Delta\omega)t)\vec{u}_x$ but $\mathbf{B_y} = 0$. Describe how the system will evolve in an *oscillating* field, compared to a *rotating* field.

Hint: Rotating fields have the form $e^{-i\omega t}$, how does this relate to an oscillating field?